

Table: Author

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Table: Project

project_id	name	deadline	start_date
1	House Price Regression	2026-04-20	2026-01-05
2	Customer Churn Classification	2026-04-22	2026-01-08
3	Image Clustering Study	2026-04-25	2026-01-12
4	Energy Forecasting	2026-04-27	2026-01-15
5	Fraud Detection	2026-04-29	2026-01-18
6	Document Topic Modeling	2026-05-02	2026-01-20

Table: PhaseComplexity

phase_name	complexity_score
Data Collection	2
Preprocessing	3
Training	5
Evaluation	3
Deployment	4

Table: Phase

phase_name	project_id	estimated_hours	description
Data Collection	1	10.0	Collect housing and sales data
Preprocessing	1	14.0	Clear missing values and scale features
Training	1	20.0	Train regression models
Evaluation	1	8.0	Evaluate MAE and R2
Data Collection	2	8.0	Gather customer churn records
Preprocessing	2	12.0	Balance classes and normalize data
Training	2	18.0	Train classification models
Evaluation	2	9.0	Compare accuracy and F1
Data Collection	3	7.0	Collect image embeddings
Preprocessing	3	10.0	Normalize clustering features
Training	3	16.0	Run clustering algorithms
Evaluation	3	7.0	Check silhouette scores
Data Collection	4	9.0	Collect energy consumption history
Preprocessing	4	13.0	Build forecasting features
Training	4	17.0	Train forecasting models
Evaluation	4	8.0	Evaluate MAE and RMSE
Data Collection	5	11.0	Collect fraud transaction logs
Preprocessing	5	15.0	Remove anomalies and encode features
Training	5	21.0	Train fraud detection classifiers
Evaluation	5	10.0	Evaluate precision and F1
Data Collection	6	6.0	Collect text corpora
Preprocessing	6	9.0	Convert documents to embeddings
Training	6	14.0	Run topic clustering
Evaluation	6	6.0	Evaluate topic coherence

Table: Hyperparameter

parameter_id	hyperparam_name	default_value	is_required	datatype
1	learning_rate	0.01	Y	FLOAT
2	batch_size	32	Y	INTEGER
3	epochs	10	Y	INTEGER
4	dropout	0.2	N	FLOAT
5	max_depth	5	N	INTEGER
6	decision_threshold	0.5	N	FLOAT
7	num_clusters	8	N	INTEGER

Table: Dataset

dataset_id	format	row_count	feature_count	creation_date
1	CSV	12000	24	2026-01-01
2	CSV	18000	31	2026-01-03
3	PARQUET	26000	42	2026-01-05
4	IMAGES	9000	128	2026-01-07
5	CSV	21000	18	2026-01-09
6	JSON	15000	36	2026-01-11

Table: HardwareConfiguration

config_id	gpu_type	cpu_cores	memory_GB
1	RX 7900	8	16.0
2	NVIDIA H100	16	32.0
3	NVIDIA A100	32	64.0
4	RTX 4090	24	128.0
5	RTX 5090	32	256.0

Table: Model

model_id	file_path	format	size_bytes
1	/models/reg_house_v1	MODEL	1000
2	/models/reg_house_v2	MODEL	1100
3	/models/forecast_v1	MODEL	1200
4	/models/churn_cls_v1	MODEL	1300
5	/models/churn_cls_v2	MODEL	1400
6	/models/fraud_cls_v1	MODEL	1500
7	/models/image_clust_v1	MODEL	1600
8	/models/image_clust_v2	MODEL	1700
9	/models/topic_clust_v1	MODEL	1800

Table: Regression

model_id	target_transform
1	log
2	none
3	standardize

Table: Classification

model_id	decision_threshold
4.0	0.5
5.0	0.55
6.0	0.6

Table: Clustering

model_id	algorithm_family
7	K-Means
8	DBSCAN
9	Hierarchical

Table: MetricType

metric_name	unit	goal	min_value	max_value
mae	score	MIN	0.0	100.0
mse	score	MIN	0.0	1000.0
rmse	score	MIN	0.0	100.0
r2	score	MAX	0.0	1.0
loss	score	MIN	0.0	100.0
accuracy	score	MAX	0.0	1.0
f1	score	MAX	0.0	1.0
silhouette	score	MAX	-1.0	1.0

Table: Run

run_id	start_time	end_time	execution_status	project_id	model_id	dataset_id	config_id
1	2026-01-15 09:20:00	2026-01-15 09:45:00	COMPLETED	1	1	1	1
2	2026-01-16 10:20:00	2026-01-16 10:50:00	COMPLETED	1	2	2	2
3	2026-01-17 11:20:00	2026-01-17 11:40:00	COMPLETED	2	4	2	2
4	2026-01-17 12:20:00	2026-01-17 15:40:00	COMPLETED	2	4	2	2
5	2026-01-26 12:20:00	2026-01-26 12:47:00	COMPLETED	2	2	2	2
6	2026-01-18 11:20:00	2026-01-18 12:05:00	FAILED	2	5	3	3
7	2026-01-19 13:20:00	2026-01-19 13:48:00	COMPLETED	3	7	4	1
8	2026-01-20 14:10:00		RUNNING	3	8	4	4
9	2026-01-21 09:20:00	2026-01-21 10:05:00	COMPLETED	4	3	5	3
10	2026-01-22 15:20:00	2026-01-22 15:42:00	COMPLETED	5	6	3	2
11	2026-01-23 16:20:00	2026-01-23 16:56:00	COMPLETED	5	4	6	4
12	2026-01-24 08:20:00	2026-01-24 09:20:00	FAILED	6	9	6	1
13	2026-01-25 10:20:00	2026-01-25 11:05:00	COMPLETED	4	1	5	5
14	2026-01-24 08:20:00	2026-01-26 12:47:00	COMPLETED	2	3	9	2

Table: Metric (Page 1/2)

metric_id	metric_name	run_id	metric_value
1	mae	1	8.9
2	mse	1	70.0
3	rmse	1	8.4
4	r2	1	0.81
5	loss	1	0.35
6	mae	2	7.8
7	mse	2	62.0
8	rmse	2	7.9
9	r2	2	0.84
10	loss	2	0.28
11	accuracy	3	0.89
12	f1	3	0.86
13	loss	3	0.42
14	accuracy	4	0.76
15	f1	4	0.72
16	loss	4	0.95
17	silhouette	5	0.61
18	loss	5	0.33
19	accuracy	6	0.91
20	f1	6	0.9
21	loss	6	0.27
22	mae	7	6.8
23	mse	7	55.0
24	rmse	7	7.4
25	r2	7	0.88

Table: Metric (Page 2/2)

metric_id	metric_name	run_id	metric_value
26	loss	7	0.22
27	accuracy	8	0.87
28	f1	8	0.84
29	loss	8	0.31
30	accuracy	9	0.93
31	f1	9	0.91
32	loss	9	0.24
33	silhouette	10	0.58
34	loss	10	0.49
35	mae	11	6.3
36	mse	11	49.0
37	rmse	11	7.0
38	r2	11	0.9
39	loss	11	0.2
40	silhouette	12	0.66
41	loss	12	0.29

Table: Creates

email	project_id
mark@email.com	1
adi@email.com	1
chaitanya@email.com	2
adi@email.com	2
john@email.com	3
bob@email.com	3
adi@email.com	4
frank@email.com	4
john@email.com	5
chaitanya@email.com	5
bob@email.com	6

Table: Uses (Page 1/2)

run_id	parameter_id	hyperparam_value
1	1	0.010
1	2	32
1	3	12
1	4	0.15
2	1	0.008
2	2	64
2	3	16
2	4	0.20
2	5	8
3	1	0.005
3	2	128
3	3	10
3	6	0.50
4	1	0.003
4	2	64
4	6	0.55
5	2	32
5	3	20
5	7	6
6	1	0.020
6	2	32
6	3	15
6	7	8
7	1	0.006
7	3	18

Table: Uses (Page 2/2)

run_id	parameter_id	hyperparam_value
7	5	10
8	1	0.002
8	2	256
8	3	8
8	6	0.62
9	1	0.004
9	2	128
9	3	14
9	4	0.10
9	6	0.58
10	2	64
10	3	9
10	7	12
11	1	0.007
11	2	32
11	3	14
11	5	7
12	1	0.009
12	2	48
12	3	11
12	7	10